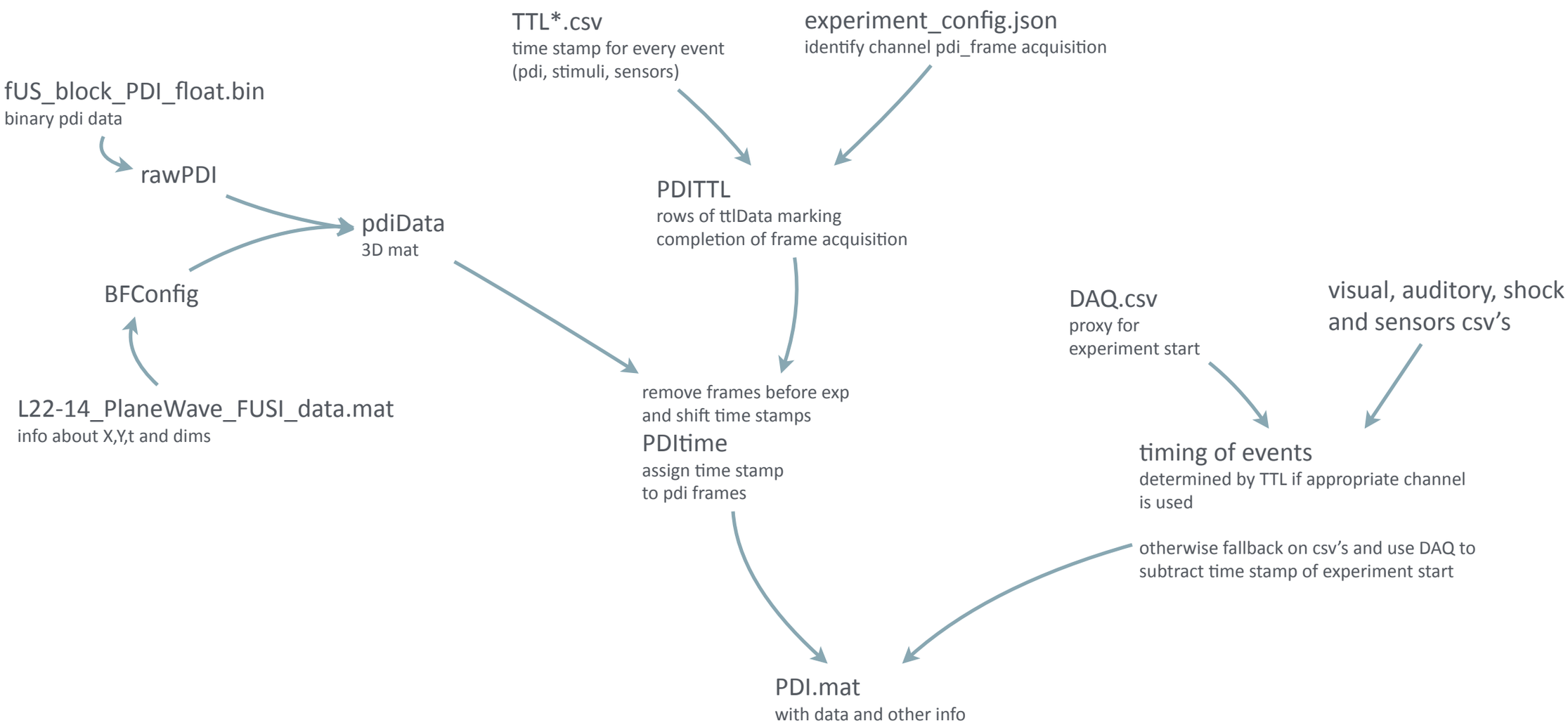
TTL*.csv
time stamp for every event
(pdi, stimuli, sensors)
experiment_config.json
identify channel pdi_frame acquisition
fUS_block_PDI_float.bin
binary pdi data
rawPDI
PDITTL
rows of ttlData marking
completion of frame acquisition
pdiData
3D mat
BFConfig
DAQ.csv
proxy for
experiment start
visual, auditory, shock
and sensors csv's
L22-14_PlaneWave_FUSI_data.mat
info about X,Y,t and dims
remove frames before exp
and shift time stamps
PDItime
assign time stamp
to pdi frames
timing of events
determined by TTL if appropriate channel
is used
otherwise fallback on csv's and use DAQ to
subtract time stamp of experiment start
PDI.mat
with data and other info

DAQ.csv

	A	B	C	D	E	F	G	H	I
1	time	visual	audio	shock_left	shock_right	shock_tail	recording_start	stim_start	camera_ttl_trigger
2	1702637451.57563	0	0	1	1	1	0	0	0
3	1702637451.57563	0	0	1	1	1	0	0	0
4	1702637463.19438	0	0	1	1	1	0	1	1
5	1702637463.20535	0	0	1	1	1	0	0	1
6	1702637583.21739	1	0	1	1	1	0	0	1
7	1702637598.20012	0	0	1	1	1	0	0	1
8	1702637643.2082	1	0	1	1	1	0	0	1
9	1702637658.21268	0	0	1	1	1	0	0	1
10	1702637703.19373	1	0	1	1	1	0	0	1
11	1702637718.20773	0	0	1	1	1	0	0	1
12	1702637763.19493	1	0	1	1	1	0	0	1
13	1702637778.19716	0	0	1	1	1	0	0	1
14	1702637823.20051	1	0	1	1	1	0	0	1
15	1702637838.20534	0	0	1	1	1	0	0	1
16	1702637883.20701	1	0	1	1	1	0	0	1
17	1702637898.19484	0	0	1	1	1	0	0	1
18	1702637943.21178	1	0	1	1	1	0	0	1
19	1702637958.20005	0	0	1	1	1	0	0	1
20	1702638003.2109	1	0	1	1	1	0	0	1
21	1702638018.19991	0	0	1	1	1	0	0	1
22	1702638063.20713	1	0	1	1	1	0	0	1
23	1702638078.20913	0	0	1	1	1	0	0	1
24	1702638123.21645	1	0	1	1	1	0	0	1
25	1702638138.21145	0	0	1	1	1	0	0	1
26	1702638189.31299	0	0	1	1	1	0	0	0

TTL.csv

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	0	0	1	1	1	0	0	0	0	0	0	1	0
2	0.0002	0	1	1	1	0	0	0	0	0	0	1	0
3	0.0004	0	1	1	1	0	0	0	0	0	0	1	0
4	0.0006	0	1	1	1	0	0	0	0	0	0	1	0
5	0.0008	0	1	1	1	0	0	0	0	0	0	1	0
6	0.001	0	1	1	1	0	0	0	0	0	0	1	0
7	0.0012	0	1	1	1	0	0	0	0	0	0	1	0
8	0.0014	0	1	1	1	0	0	0	0	0	0	1	0
9	0.0016	0	1	1	1	0	0	0	0	0	0	1	0
10	0.0018	0	1	1	1	0	0	0	0	0	0	1	0

VisualStimulation.csv

A	B	C	D
time	stim	event	duration
1702637463.19039	visual	start	0
1702637583.20343	visual	stim	15
1702637598.19812	visual	blank	0
1702637643.20521	visual	stim	15
1702637658.20271	visual	blank	0
1702637703.19074	visual	stim	15
1702637718.20474	visual	blank	0
1702637763.19294	visual	stim	15
1702637778.19317	visual	blank	0
1702637823.19552	visual	stim	15
1702637838.19536	visual	blank	0

DAQ.csv

time	visual	audio	shock_left	shock_right	shock_tail	recording_start	stim_start	camera_ttl_trigger
1702637451.57563	0	0	1	1	1	0	0	0
1702637451.57563	0	0	1	1	1	0	0	0
1702637463.19438	0	0	1	1	1	0	1	1
1702637463.20535	0	0	1	1	1	0	0	1
1702637583.21739	1	0	1	1	1	0	0	1
1702637598.20012	0	0	1	1	1	0	0	1
1702637643.2082	1	0	1	1	1	0	0	1
1702637658.21268	0	0	1	1	1	0	0	1
1702637703.19373	1	0	1	1	1	0	0	1
1702637718.20773	0	0	1	1	1	0	0	1
1702637763.19493	1	0	1	1	1	0	0	1
1702637778.19716	0	0	1	1	1	0	0	1
1702637823.20051	1	0	1	1	1	0	0	1
1702637838.20534	0	0	1	1	1	0	0	1
1702637883.20701	1	0	1	1	1	0	0	1
1702637898.19484	0	0	1	1	1	0	0	1
1702637943.21178	1	0	1	1	1	0	0	1
1702637958.20005	0	0	1	1	1	0	0	1
1702638003.2109	1	0	1	1	1	0	0	1
1702638018.19991	0	0	1	1	1	0	0	1
1702638063.20713	1	0	1	1	1	0	0	1
1702638078.20913	0	0	1	1	1	0	0	1
1702638123.21645	1	0	1	1	1	0	0	1
1702638138.21145	0	0	1	1	1	0	0	1
1702638189.31299	0	0	1	1	1	0	0	0

TTL.csv

Determining Experiment start

Both the pdiData and the time stamp of pdi frames and other events (e.g. stimuli) is set to the beginning of the experiment (experiment start). This is done according to the info in the appropriate channel in TTL.

TTL also records the timing of the stimuli and other events (e.g. wheel) in appropriate channels, however *in some cases* this info might be missing from the TTL. In these cases, the first record of the **DAQ.csv** file is taken as experiment start. This is then removed from the csv of the single instruments, e.g. VisualStimulation.csv

VisualStimulation.csv

A	B	C	D
time	stim	event	duration
1702637463.19039	visual	start	0
1702637583.20343	visual	stim	15
1702637598.19812	visual	blank	0
1702637643.20521	visual	stim	15
1702637658.20271	visual	blank	0
1702637703.19074	visual	stim	15
1702637718.20474	visual	blank	0
1702637763.19294	visual	stim	15
1702637778.19317	visual	blank	0
1702637823.19552	visual	stim	15
1702637838.19536	visual	blank	0

DAQ.csv

time	visual	audio	shock_left	shock_right	shock_tail	recording_start	stim_start	camera_ttl_trigger
1702637451.57563	0	0	1	1	1	0	0	0
1702637451.57563	0	0	1	1	1	0	0	0
1702637463.19438	0	0	1	1	1	0	1	1
1702637463.20535	0	0	1	1	1	0	0	1
1702637583.21739	1	0	1	1	1	0	0	1
1702637598.20012	0	0	1	1	1	0	0	1
1702637643.2082	1	0	1	1	1	0	0	1
1702637658.21268	0	0	1	1	1	0	0	1
1702637703.19373	1	0	1	1	1	0	0	1
1702637718.20773	0	0	1	1	1	0	0	1
1702637763.19493	1	0	1	1	1	0	0	1
1702637778.19716	0	0	1	1	1	0	0	1
1702637823.20051	1	0	1	1	1	0	0	1
1702637838.20534	0	0	1	1	1	0	0	1
1702637883.20701	1	0	1	1	1	0	0	1
1702637898.19484	0	0	1	1	1	0	0	1
1702637943.21178	1	0	1	1	1	0	0	1
1702637958.20005	0	0	1	1	1	0	0	1
1702638003.2109	1	0	1	1	1	0	0	1
1702638018.19991	0	0	1	1	1	0	0	1
1702638063.20713	1	0	1	1	1	0	0	1
1702638078.20913	0	0	1	1	1	0	0	1
1702638123.21645	1	0	1	1	1	0	0	1
1702638138.21145	0	0	1	1	1	0	0	1
1702638189.31299	0	0	1	1	1	0	0	0